Hochschule Bremen
City University of Applied Sciences

HSB

Comparative Analysis of Native vs. Cross-Platform Mobile Development

An Empirical Performance Study

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Outline:

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4. Methodology
5. Current Status & Roadmap

Motivation

The Developer's Dilemma

- **Cost vs. Quality:** Startups need to minimize costs (Cross-Platform) but fear compromising User Experience (Native).
- **The Problem: I**s the "Performance Gap" still relevant for modern frameworks?
- **The Goal:** Providing a data-driven decision guide for 2026.

Research Question

To what extent does the performance overhead of state-of-the-art cross-platform frameworks (Flutter) impact system resources compared to native development (Swift)?

Key Metrics:

- CPU Efficiency (Calculation speed)
- UI Rendering (FPS stability)
- Memory Footprint (RAM usage)

Related Work & Research Gap

State of the Art

- Majchrzak et al. (2017): Established taxonomy of frameworks but data is outdated.
- You & Hu (2021): Comparative study (Flutter vs. Native).
  - Result: Native was more efficient in Memory & CPU.
  - Limitation: Tested only on Android using Emulators.

The Research Gap

- Lack of iOS Data: Recent studies focus heavily on Android.
- Real-World Accuracy: Emulators do not reflect real hardware constraints.
- Contribution: Performance analysis on Physical iOS Devices.

Methodology

Experimental Design

- **Approach**: Empirical Comparative Study
- **The Artifact**: A standardized "Benchmark App"
- **Implementation**:
  1. Native: iOS (Swift)
  2. Cross-Platform: Flutter (Dart)
- **Test Environment**: Physical iPhone (iOS 26) – No Emulators.
- **Stress Test Scenarios**:
  - CPU: Prime number calculation (Sieve of Eratosthenes).
  - GPU: Rendering 1,000 high-res images.

Current Status & Roadmap

Status Quo:

- ✅ Research Gap Identified: Confirmed lack of recent benchmarks in current literature.
- ✅ Benchmark Strategy: Defined exact algorithms (Prime Sieve) and success metrics (CPU/RAM) for fair comparison.
- ✅ Technical Foundation: Verified build environments for Swift (Xcode) and Flutter.

Roadmap:

- Phase 1 (Dec - Jan W1): Implementation of Native Swift App.
- Phase 2 (Jan W2 - Jan W3): Implementation of Flutter App.
- Phase 3 (Jan W4): Measurement & Data Collection.
- Phase 4 (Feb): Evaluation & Final Writing.

Thank you for your attention!
Questions?